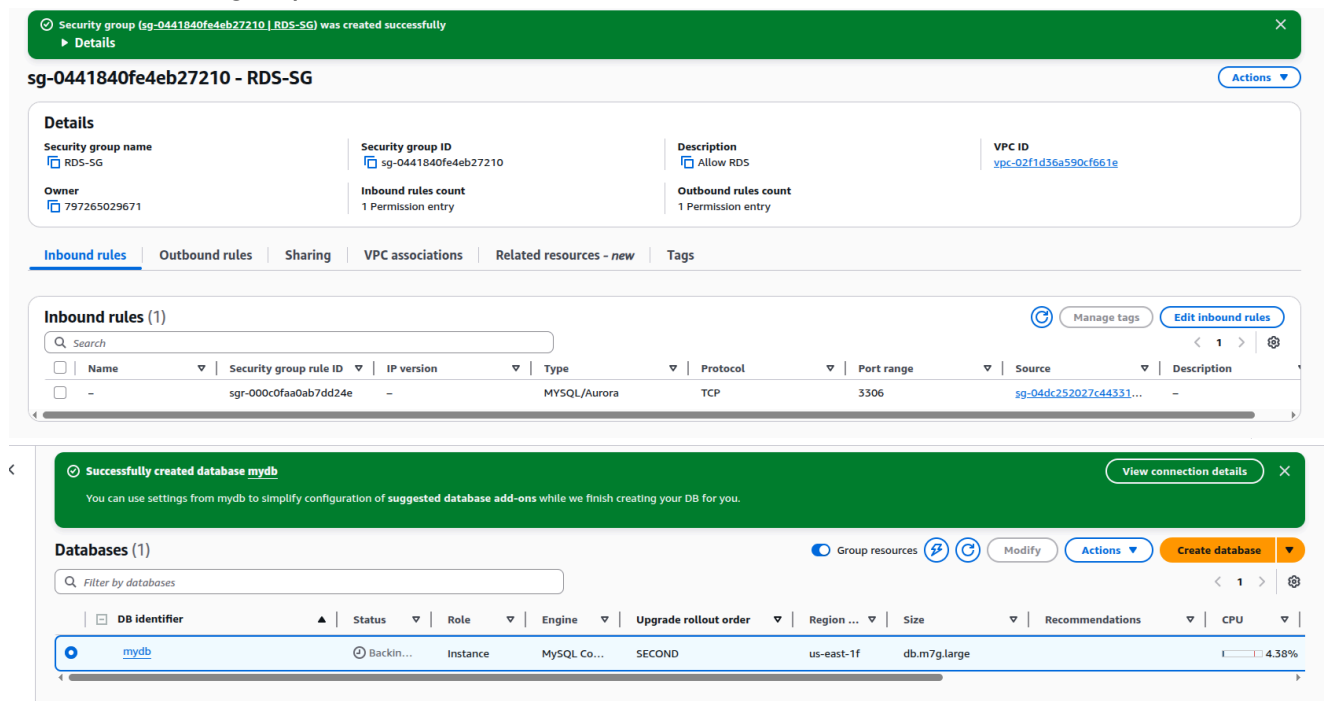


Q1. Connect an EC2 instance to an RDS instance in the same VPC using the CLI. Provide a screenshot.

Ans:

Created EC2 and RDS Instance and My VPC , with security groups created for RDS and EC2 , the RDS group SG is attached with EC2 SG as source.



The Connection succeeds .

```
ubuntu@ip-10-0-1-87: ~/certs
mysql -h mydb.cely4ams0gyn.us-east-1.rds.amazonaws.com \
-P 3306 \
-u admin \
-p'test12345' \
--ssl-mode=VERIFY_CA \
--ssl-ca=/home/ubuntu/certs/global-bundle.pem \
mysql
mysql: [Warning] Using a password on the command line interface can be insecure.
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 37
Server version: 8.4.7 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>
```

Q2. Create a read replica for a MySQL/PostgreSQL RDS instance in the same region, and promote this replica to a standalone RDS?

Ans:

First I created PostgreSQL RDS instance in one region , Now I am creating the read replica for that instance in the same region.

The screenshot shows the AWS RDS console for a PostgreSQL instance named 'myrds'. The instance is in the 'Configuring-enhanced-monitoring' status. The engine is PostgreSQL, and the region is us-east-1b. The class is db.m7g.large. The CPU usage is 0.00%. The instance is connected to the internet access gateway, which is disabled. The programming language is PSQL (macOS), and the endpoint type is Instance endpoint. The connection steps are as follows:

```
1 export RDSHOST="myrds.cely4ams0gyn.us-east-1.rds.amazonaws.com"
2 psql "host=$RDSHOST port=5432 dbname=Test user=postgres sslmode=verify-full sslrootcert=/certs/global-bundle.pem password=<Enter_DB_Password>"
```

Now my read replica is promoted to standalone instance .

The screenshot shows the AWS RDS console for a PostgreSQL instance named 'replicards'. The instance is in the 'Available' status. The engine is PostgreSQL, and the region is us-east-1f. The class is db.m7g.large. The CPU usage is 3.08%. The instance is connected to the internet access gateway, which is disabled. The programming language is PSQL (macOS), and the endpoint type is Instance endpoint. The connection steps are as follows:

```
1 export RDSHOST="replicards.cely4ams0gyn.us-east-1.rds.amazonaws.com"
2 psql "host=$RDSHOST port=5432 dbname=Test user=postgres sslmode=verify-full sslrootcert=/certs/global-bundle.pem password=<Enter_DB_Password>"
```

Q3. After inserting data into the primary RDS, how can you verify that the data is replicated to the read replica, and how can you check the replication lag? Provide a screenshot of your verification.

Ans :

Using My Primary RDS from above question which I created , Next I created an EC2 instance to check the connection , connected to my RDS then added test data in it to check .

```
ubuntu@ip-10-0-1-69:/certs$ psql "host=myrds.cely4ams0gyn.us-east-1.rds.amazonaws.com \
port=5432 \
dbname=Test \
user=postgres \
password=test12345 \
sslmode=verify-full \
sslrootcert=/certs/global-bundle.pem"
psql (16.11 (Ubuntu 16.11-0ubuntu0.24.04.1), server 17.6)
WARNING: psql major version 16, server major version 17.
         Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
Type "help" for help.

Test=> CREATE TABLE replication_test(id Serial Primary Key , message TEXT , created_At TIMESTAMP DEFAULT current_timestamp);
CREATE TABLE
Test=> Insert into replication_test(message) values('Hello testing from database');
INSERT 0 1
Test=> select * from replication_test;
 id |          message          |          created_at
-----+-----+-----
  1 | Hello testing from database | 2026-01-31 05:58:50.375651
(1 row)

Test=> |
```

The read replica works well when checked

```
Test=> select * from replication_test;
 id |          message          |          created_at
-----+-----+-----
  1 | Hello testing from database | 2026-01-31 05:58:50.375651
(1 row)

Test=> |
```

Q4. Create a read-only user and connect to the read replica, how can you verify that SELECT queries are actually served by the replica? Provide a screenshot.

Ans:

Connected to My Primary RDS instance and then created a sample data on it .

```
ubuntu@ip-10-0-1-69: ~  
Some psql features might not work.  
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression:  
off)  
Type "help" for help.  
Test=> CREATE TABLE public.employee (  
    id SERIAL PRIMARY KEY,  
    name TEXT,  
    role TEXT  
);  
  
INSERT INTO public.employee (name, role)  
VALUES ('Mukesh', 'Engineer'), ('Piyush', 'DevOps');  
  
SELECT * FROM public.employee;  
CREATE TABLE  
INSERT 0 2  
 id | name | role  
-----  
  1 | Mukesh | Engineer  
  2 | Piyush | DevOps  
(2 rows)  
  
Test=> 
```

Now creating a read-only user in my primary RDS.

```
ubuntu@ip-10-0-1-69: ~  
 2 | Piyush | DevOps  
(2 rows)  
  
Test=> CREATE USER readonly_user WITH PASSWORD 'readonly123';  
CREATE ROLE  
Test=> GRANT CONNECT ON DATABASE Test TO readonly_user;  
GRANT USAGE ON SCHEMA public TO readonly_user;  
GRANT SELECT ON ALL TABLES IN SCHEMA public TO readonly_user;  
ERROR: database "test" does not exist  
GRANT  
GRANT  
Test=> GRANT CONNECT ON DATABASE public TO readonly_user;  
GRANT USAGE ON SCHEMA public TO readonly_user;  
GRANT SELECT ON ALL TABLES IN SCHEMA public TO readonly_user;  
ERROR: database "public" does not exist  
GRANT  
GRANT  
Test=> GRANT CONNECT ON DATABASE "Test" TO readonly_user;  
GRANT USAGE ON SCHEMA public TO readonly_user;  
GRANT SELECT ON ALL TABLES IN SCHEMA public TO readonly_user;  
GRANT  
GRANT  
GRANT  
Test=> 
```

Now this is my read replica which serves read query only and when I tried to insert data into

it,



The screenshot shows a terminal window with the title bar 'ubuntu@ip-10-0-1-69: ~'. The terminal contains the following text:

```
Type "help" for help.

Test=> SELECT current_database();
SHOW transaction_read_only;
SELECT * FROM public.employee;
current_database
-----
Test
(1 row)

transaction_read_only
-----
on
(1 row)

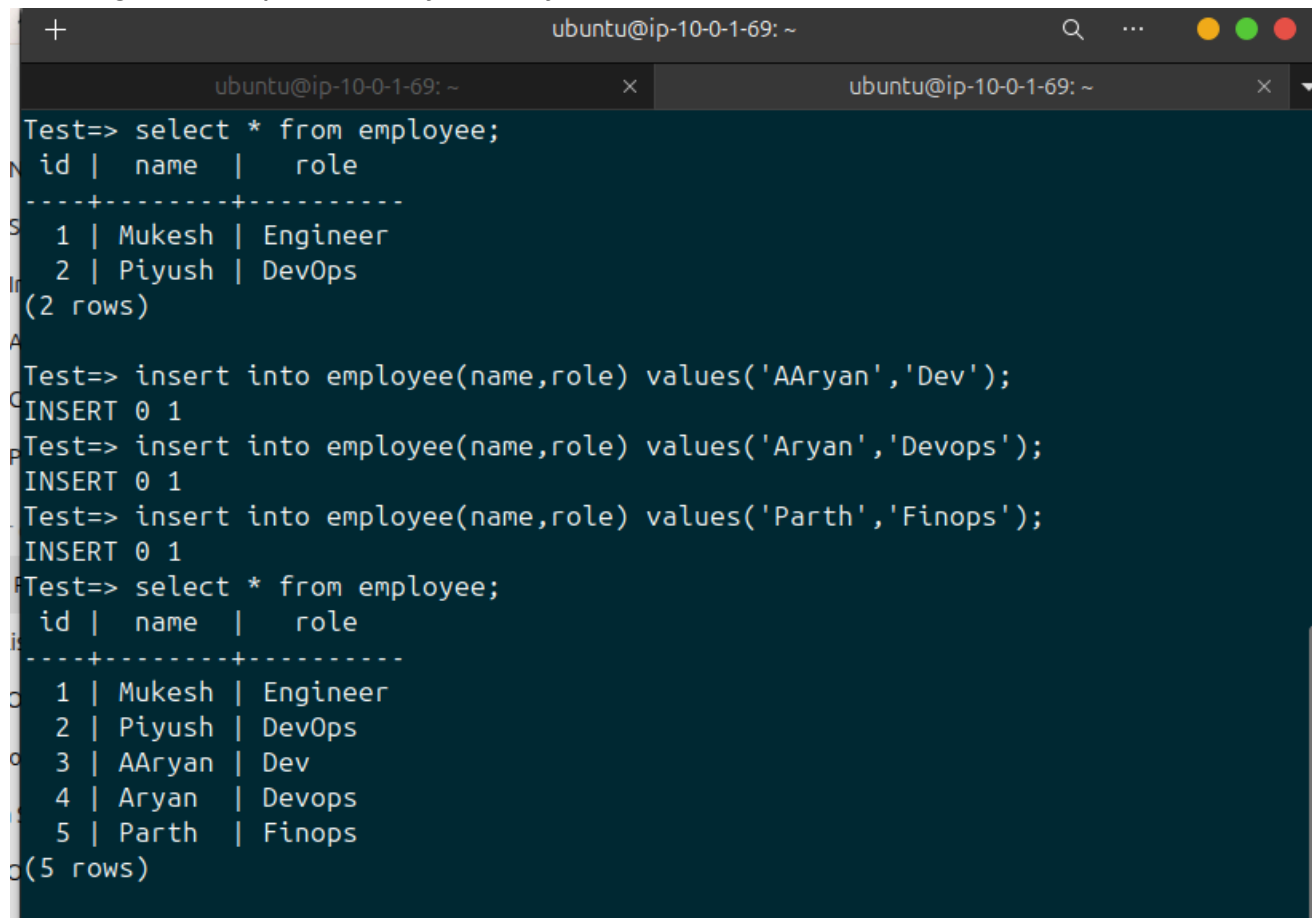
id | name  | role
---+----+---
 1 | Mukesh | Engineer
 2 | Piyush | DevOps
(2 rows)

Test=> Insert into public.employee values(3,'Raj','QA');
ERROR:  cannot execute INSERT in a read-only transaction
Test=> 
```

Q5. Run 10 SELECT queries on the read replica and 5 INSERT queries on the primary RDS from an EC2 instance, how can you monitor which instance is serving which queries? Provide a screenshot.

Ans:

Running 5 insert queries in my Primary RDS.



```
+ ubuntu@ip-10-0-1-69: ~
Test=> select * from employee;
id | name | role
----+-----+-----
1 | Mukesh | Engineer
2 | Piyush | DevOps
(2 rows)

Test=> insert into employee(name,role) values('AAryan','Dev');
INSERT 0 1
Test=> insert into employee(name,role) values('Aryan','Devops');
INSERT 0 1
Test=> insert into employee(name,role) values('Parth','Finops');
INSERT 0 1
Test=> select * from employee;
id | name | role
----+-----+-----
1 | Mukesh | Engineer
2 | Piyush | DevOps
3 | AAryan | Dev
4 | Aryan | Devops
5 | Parth | Finops
(5 rows)
```

Now running 10 select queries in read replica

```

WARNING: psql major version 16, server major version 17.
Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
Type "help" for help.

Test=> select * from employee;
id | name | role
---+-----+-----
 1 | Mukesh | Engineer
 2 | Piyush | DevOps
 3 | AAryan | Dev
 4 | Aryan | Devops
 5 | Parth | Finops
(5 rows)

Test=> select * from employee where id =1;
id | name | role
---+-----+-----
 1 | Mukesh | Engineer
(1 row)

Test=> select * from employee where id =2;
id | name | role
---+-----+-----
 2 | Piyush | DevOps
(1 row)

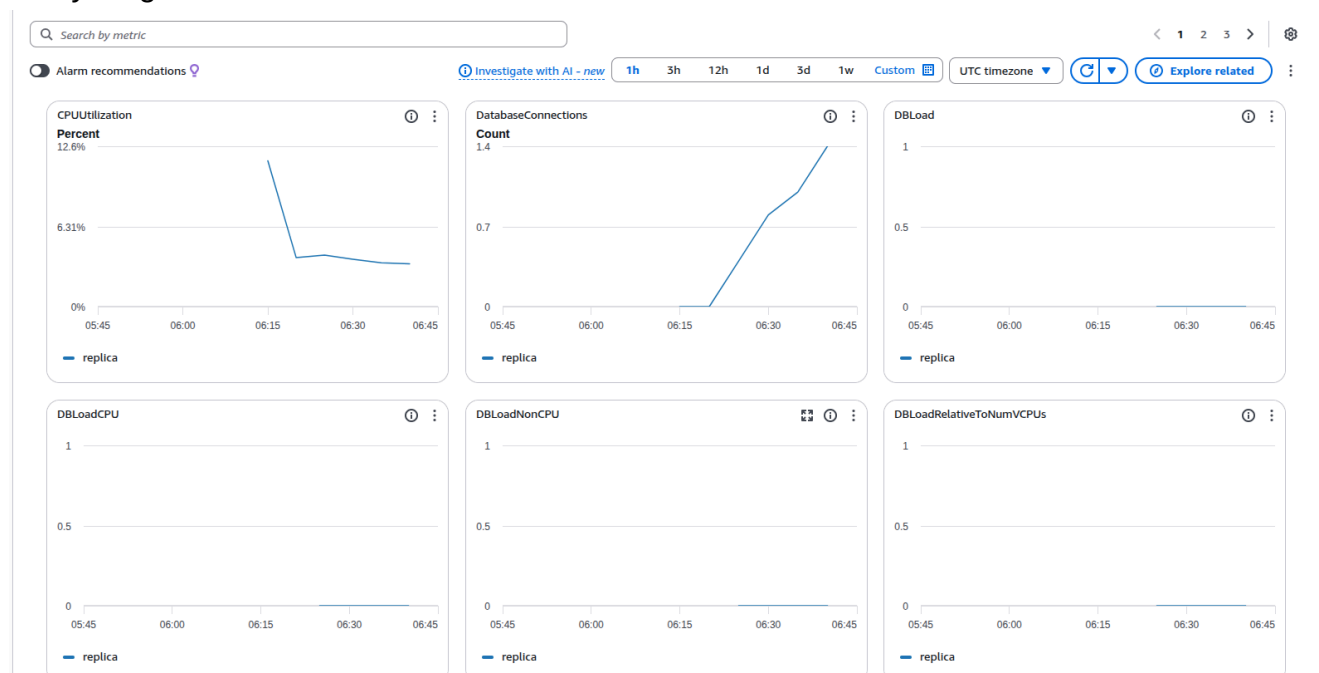
Test=> select count(*) from employee;
count
-----
      5
(1 row)

Test=> select name from employee where role ='Dev';
name
-----
AAryan
(1 row)

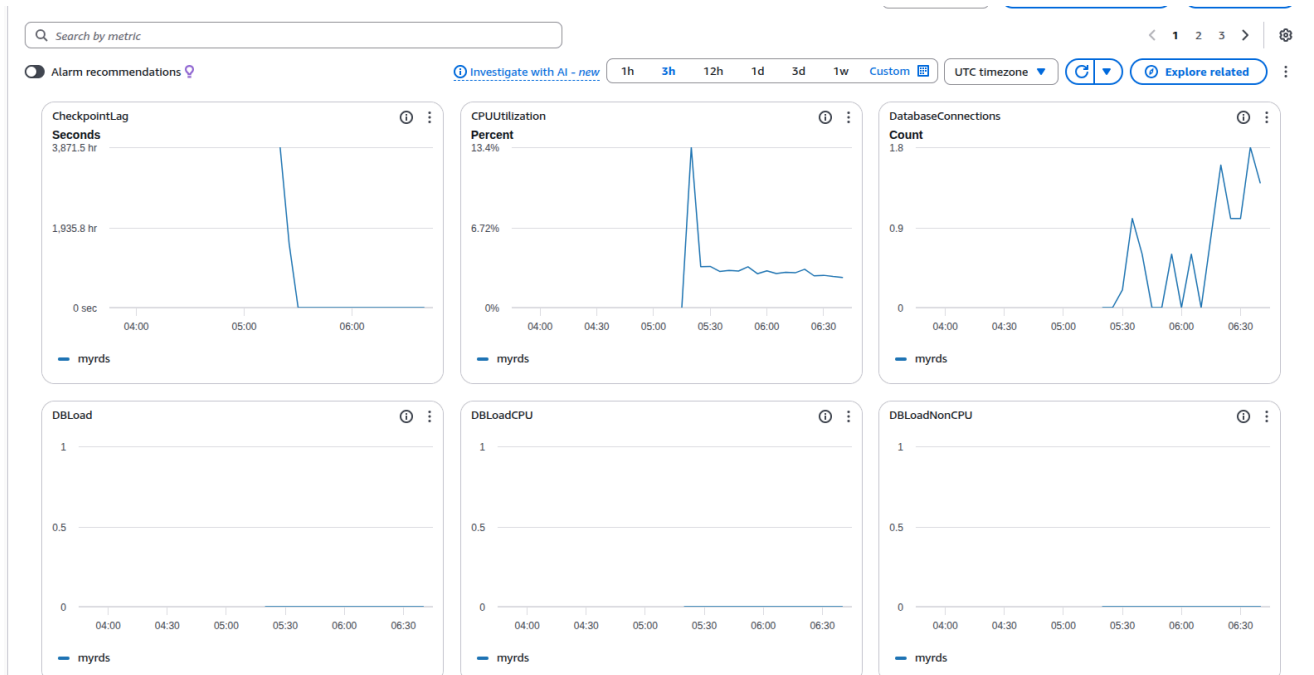
Test=> select * from employee order by id desc;
id | name | role
---+-----+-----
 5 | Parth | Finops
 4 | Aryan | Devops
 3 | AAryan | Dev
 2 | Piyush | DevOps
 1 | Mukesh | Engineer
(5 rows)

```

Using Cloudwatch metrics in monitoring in read replica rds which shows dbcount , dbload everything.



This is the cloudwatch metrics for my Primary RDS.



Q6. Connect an EC2 instance to RDS using an IAM authentication token, and how can you verify that the connection is successful? Provide a screenshot.

Ans:

Updated my Primary RDS and added IAM authentication with it .

The screenshot displays the AWS RDS console for the Primary RDS instance 'myrds'. The instance is in the 'Available' state and has IAM Authentication enabled. The console shows the 'Connect using' section with 'Code snippets' selected, and the 'IAM authentication token' section with 'Get token' button. The 'Connection steps' section shows the commands to connect to the RDS instance using the IAM authentication token.

Related

DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU	Current...	Mainte...
myrds	Available	Primary	PostgreSQL	us-east-1b	db.m7g.la...	1 Informational	2.92%	0.00 se	none
replica	Available	Replica	PostgreSQL	us-east-1b	db.m7g.la...		3.15%	0.00 se	none

Connect using | Info

- Code snippets** (Selected): Use when connecting through SDK, APIs, or third-party tools including agents.
- CloudShell**: Use for a quick access to AWS CLI that launches directly from the AWS Management Console.
- Endpoints**: Use when connecting through any IDE interface.

Internet access gateway
Disabled

IAM authentication token
Get token

Programming language
PSQL (macOS)

Endpoint type
Instance endpoint

Connection steps
Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

```
1 export RDSHOST="myrds.cely4ams0gyn.us-east-1.rds.amazonaws.com"
2 psql "host=$RDSHOST port=5432 dbname=Test user=postgres sslmode=verify-full sslrootcert=/certs/global-bundle.pem password=$(aws rds generate-db-auth-token --hostname
```


Successfully attached EC2-RDS-IAM-Role to instance i-0956653d7819d17fb
✕

Instances (1/1) Info

All states ▾
< 1 >
⚙️

☒	Name 🔗 ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status	Availability Zone ▾	Public IPv4 DNS ▾	Public IPv4 ... ▾	Elastic IP
☒	RDSEC2	i-0956653d7819d17fb	✔️ Running 🔗 🔗	t3.micro	✔️ 3/3 checks passed View alarms +		us-east-1f	-	44.197.112.165	-

i-0956653d7819d17fb (RDSEC2) ⚙️ ▾

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary Info

Instance ID
🔗 i-0956653d7819d17fb

IPv6 address
 -

Hostname type
 IP name: ip-10-0-1-69.ec2.internal

Answer private resource DNS name
 -

Auto-assigned IP address
🔗 44.197.112.165 [Public IP]

IAM Role
🔗 EC2-RDS-IAM-Role 🔗

IMDSv2
 Required

Public IPv4 address
🔗 44.197.112.165 | open address 🔗

Instance state
✔️ Running

Private IP DNS name (IPv4 only)
🔗 ip-10-0-1-69.ec2.internal

Instance type
t3.micro

VPC ID
🔗 vpc-02f1d36a590c661e (MyVPC) 🔗

Subnet ID
🔗 subnet-0938683593c34b5c8 (My-Subnet) 🔗

Instance ARN
🔗 arn:aws:ec2:us-east-1:797265029671:instance/i-0956653d7819d17fb

Private IPv4 addresses
🔗 10.0.1.69

Public DNS
 -

Elastic IP addresses
 -

AWS Compute Optimizer finding
🔗 Opt-in to AWS Compute Optimizer for recommendations. | Learn more 🔗

Auto Scaling Group name
 -

Managed
 false

```
+ ubuntu@ip-10-0-1-69: ~  
ubuntu@ip-10-0-1-69:~$ export RDSHOST="myrds.cely4ams0gyn.us-east-1.rds.amazonaws.com"  
export REGION="us-east-1"  
export DBUSER="db_iam_user"  
  
TOKEN=$(aws rds generate-db-auth-token \  
--hostname $RDSHOST \  
--port 5432 \  
--region $REGION \  
--username $DBUSER)  
ubuntu@ip-10-0-1-69:~$ echo $TOKEN  
myrds.cely4ams0gyn.us-east-1.rds.amazonaws.com:5432/?Action=connect&DBUser=db_ia  
m_user&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=ASIA3TIFS5IT2EUZ235Z%2F  
20260131%2Fus-east-1%2Frds-db%2Faws4_request&X-Amz-Date=20260131T074316Z&X-Amz-E  
xpires=900&X-Amz-SignedHeaders=host&X-Amz-Security-Token=IQoJb3JpZ2luX2VjE0j%2F%  
2F%2F%2F%2F%2F%2F%2F%2FwEaCXVzLWVhc3QtMSJIMEYCIQCNH3DatN%2Fro5cSWiLNfL%2BRjpa  
qcIUo6wHQicj1TR9P0giHaOcmldmC%2BnsERWylzObcoiWvmKF6069NTXLDC%2F3hp6KKsIFCLH%2F%  
2F%2F%2F%2F%2F%2F%2F%2FwEQABOMnzk3MjY1MDI5NjcXlgzjoArbDsLo2S3W9mAqlgXldqD530w  
T4eWaGw3f5qHuP7pvXODceApmt8N%2FVYllrPHZfjEYdZIF6EjUrruzfOp23cca02NtQd7BqRh7grKQ  
suLh%2Bq500YAbaCiFtJGMvCnzec0Ag%2FaRgoJ6mA%2FJYDYBDwkjGuXF8qmZ7Llbc0ZVuSVHQEL8T44  
xA69desLLLeJpmXWOA0CcCo1eza%2FaH1z3lBlVjtQWq2Ia8uNdrr1xsKa8BPrixWVP71hiMs07GI1S  
jspdoIu6ufQxWN2lVPYYbBhY9uydJG0uXkMZil%2BRzf10oZwjrmek3vQTzsWGbdqQo3ExVQJ3eOCsN  
pzTQ24d6IOy9qbJtcYaNPfvNF%2FjvgJ2z2jldpQSzr0%2Fbb0WBIPVz0NEZlsBAVT1H5qLcrAML4Vab  
UD7QKDraMyjSsgtmbZR1MPC%2B5wHewxgKM%2BXz5BLjHcx3BBiF5vEts004K0IACgxPxCaGDHbjDp3e  
btC0jdFI3xLnXWZMt%2BozTPfOMEcqvtPMZpFTQLmroBXLjhJYg1Es9JIUN464flkdq8WPt0QsqCb70  
%2BklMJ5pVa95DvXVJEPWUL81A%2BndnzyFPossalHhZocBIfpdT6phf0ZxxfWMDATVL3Gb5X%2BiMRc  
L7U6C736p61uLp32NLti0iTMOT-GUHLDPX-hBN-u4HUUA7-z3E2-tN-i0GG0GCUH-Z-KFY
```

and it logs into the RDS

```
postgres
ubuntu@ip-10-0-1-69:~$ psql "host=$RDSHOST port=5432 dbname=Test user=db_iam_user
sslmode=verify-full sslrootcert=/certs/global-bundle.pem password=$(aws rds ge
nerate-db-auth-token --hostname $RDSHOST --port 5432 --username db_iam_user --re
gion us-east-1)"
psql (16.11 (Ubuntu 16.11-0ubuntu0.24.04.1), server 17.6)
WARNING: psql major version 16, server major version 17.
        Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression:
off)
Type "help" for help.

Test=> select current_user;
 current_user
-----
 db_iam_user
(1 row)

Test=>
```

Q7. Test Multi-AZ failover on a primary RDS instance, and in what ways is a read replica different from a Multi-AZ deployment?

Ans:

Created My Multi-AZ RDS and then first checked the ip address in it.

```
+ ubuntu@ip-10-0-1-69: ~
piyush@CK-piyush: ~ x ubuntu@ip-10-0-1-69: ~ x
ubuntu@ip-10-0-1-69:~$ export RDSHOST="myrds.cely4ams0gyn.us-east-1.rds.amazonaw
s.com"
psql "host=$RDSHOST port=5432 dbname=Test user=postgres sslmode=verify-full sslr
ootcert=/certs/global-bundle.pem password=test12345"
psql (16.11 (Ubuntu 16.11-0ubuntu0.24.04.1), server 17.6)
WARNING: psql major version 16, server major version 17.
        Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression:
off)
Type "help" for help.

Test=> select inet_server_addr();
 inet_server_addr
-----
 10.0.2.71
(1 row)

Test=>
```

After the failover the IP changed for primary RDS

```
Test=> \q
ubuntu@ip-10-0-1-69:~$ psql "host=$RDSHOST port=5432 dbname=Test user=postgres s
sslmode=verify-full sslrootcert=/certs/global-bundle.pem password=test12345"
psql (16.11 (Ubuntu 16.11-0ubuntu0.24.04.1), server 17.6)
WARNING: psql major version 16, server major version 17.
         Some psql features might not work.
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression:
off)
Type "help" for help.

Test=> select inet_server_addr();
 inet_server_addr
-----
 10.0.1.84
(1 row)

Test=> 
```

Multi-AZ provides high availability by replicating the data in synchronous manner to a instance which is standby and will automatically fails during the outages whereas Read Replicas provides the availability asynchronously by replicating data in asynchronous manner from the primary databases and then are used to scale workloads.